D 10243 **(Pages : 4)** **Name.....**

Reg. No.....

FIFTH SEMESTER B.A./B.Sc. DEGREE EXAMINATION, NOVEMBER 2021

(CUCBCSS—UG)

Physics/Applied Physics

PHY 5B 06/APY 5B 07—ELECTRODYNAMICS—II

Time : Three Hours Maximum : 80 Marks

The symbols used in the question paper have their usual meanings.

Section A

Answer in a word **or** *a phrase.*
Answer **all** *questions.*
Each question carries 1 *mark.*

1. Magnetic field can be produced by a current or by a changing ————.
2. The average power per unit area transported by an electromagnetic wave is called its ————.
3. The smaller is the time constant CR, the ———— is the discharge of the capacitor in a series CR circuit.
4. The ratio of r.m.s. value of current to the mean value of current is called ————.
5. Higher the quality factor of a circuit, ———— is its bandwidth.

Write True *or* False :

6. Electromagnetic waves travel with the speed of light in vacuum.
7. The tangential component of H is continuous across the boundary between two media.
8. If the value of L/R in an LR series circuit increases, the time taken by the current to reach its maximum value decreases.
9. In pure inductive circuits, the current is lagging behind the e.m.f. by $\pi/2$ in phase.
10. Thevenin's theorem can be applied to networks with DC only.

(10 × 1 = 10 marks)

Turn over

Section B

Answer **all** *questions in two* **or** *three sentences.*
Each question carries 2 *marks.*

11. State and explain Faraday's law in electromagnetic induction.

12. Write down the importance of displacement current in Maxwell's equations.

13. Define Poynting vector and give an expression for the same.

14. Give an explanation for ballistic galvanometer. What are the conditions for a moving coil galvanometer to be ballistic ?

15. Give and expression for the instantaneous current in a series LR circuit. What are the terms involved ?

16. Draw the basic circuit of an AC bridge and write down the condition to balance it.

17. State and explain maximum power transfer theorem.

(7 × 2 = 14 marks)

Section C

Answer any **five** *questions in* **a paragraph** *of about* **half a page** *to* **one page**.
Each question carries 4 *marks.*

18. Explain briefly about magnetic charge.

19. Obtain an expression for the energy stored in a magnetic field in terms of current.

20. Show that the energy flux density transported by the field is given by $\frac{1}{\mu_0} E \times B$.

21. Obtain an expression for the growth and decay of charge in a capacitor through a resistor.

22. Obtain equations for r.m.s. value of e.m.f. and current in an AC circuit.

23. What are the basic steps for converting a voltage source with a series resistance into an equivalent current source with a parallel resistance ?

24. State Norton's theorem. Give the different steps involved in Nortanizing a given circuit network.

(5 × 4 = 20 marks)

Section D

Problems-Write all relevant formulas, all important steps carry separate marks.
Answer any **four** *questions.*
Each question carries 4 *marks.*

25. Write down Neumann's formula for mutual induction. Explain its importance.

26. Find the magnetic flux through a solenoid of length *l* with number of turns per unit length N and radius R carrying a current I. Also calculate self inductance per unit length of the coil.

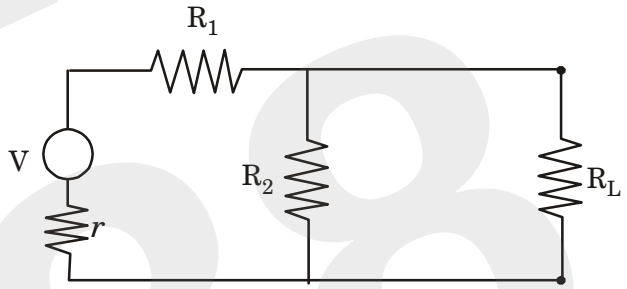
27. Find the energy stored in a long coaxial cable of length '*l*' carries a current 'I' flowing down the surface of the inner cylinder of radius '*a*' and back along the outer cylinder of radius '*b*'.

28. Write down the boundary conditions for the electric and magnetic vectors E and B at an interface separating two linear dielectrics of permittivities ε_1 and ε_2 and permeabilities μ_1 and μ_2.

29. A capacitor of 0.2 μF is first charged and discharged through a resistance of 10 megaohms. Find the time, the potential takes to fall to ¼ of its original value.

30. An alternating voltage of 100 V at a frequency of 25 Hz is applied to a circuit consisting a resistance 1.5 Ω and an inductance of 0.01 Henry in series : (a) Find the current flowing ; (b) Phase difference between e.m.f. and current ; and (c) Potential drop across resistor and inductor.

31. Apply Thevenin's theorem to find the current through the load resistance, RL = 15 Ω in the following network. Given R_1 = 3 Ω, R_2 = 12 Ω, V = 24 Volts with an internal resistance r = 1 Ω :



(4 × 4 = 16 marks)

Turn over

Section E (Essays)

Answer in about **two pages** *Answer any* **two** *questions.*
Each question carries 10 *marks.*

32. Derive the Maxwell's equations inside a polarized matter.

33. Find expressions for the transmission and reflection coefficients when a plane polarized monochromatic wave of angular frequency ω passes normally through the boundary between two linear dielectrics.

34. Discuss the growth of current in a circuit containing an inductance L and a resistance R connected in series with a cell of steady e.m.f. Explain the time constant of the circuit.

35. Obtain expressions for resultant e.m.f., impedence and power factor of an LCR series circuit when an alternating current is flowing through it. Explain the resonance in LCR series circuit.

(2 × 10 = 20 marks)